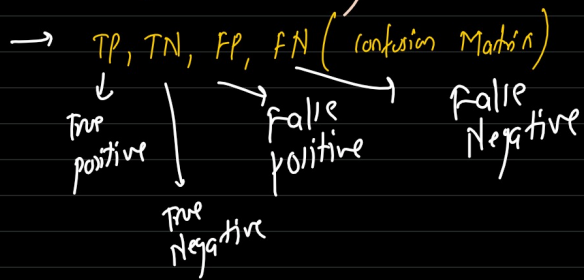


31-05-2026

Agenda: →

- Evaluation Metric (classification)
- Equation (Linear, Polynomial)
- Gradient Descent

Evaluation Metric
(classification)

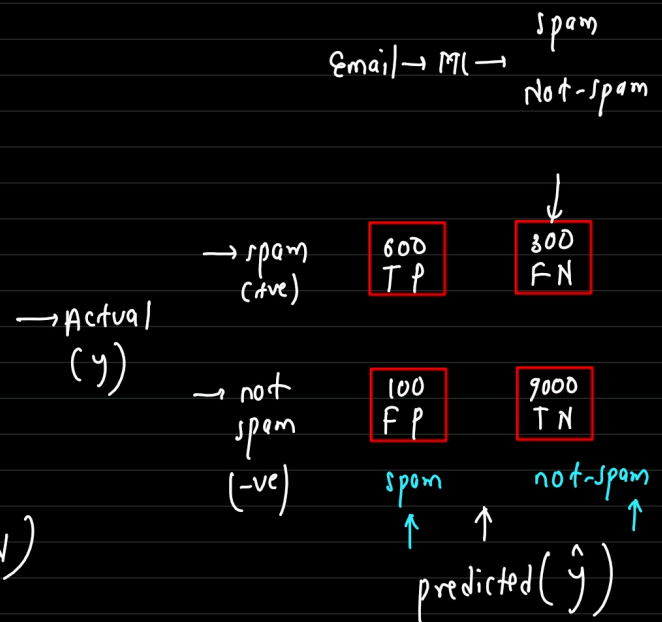


$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$
$$\text{Precision} = \frac{(TP)}{(TP + FP)}$$

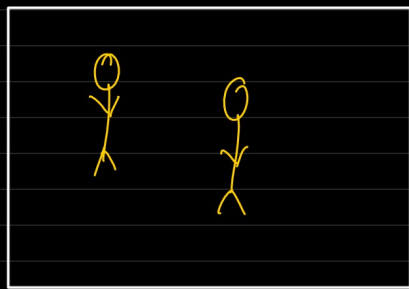
considers -ve prediction accuracy

$$\text{Recall} = \frac{(TP)}{(TP + FN)}$$

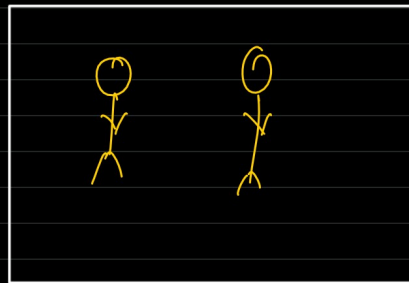
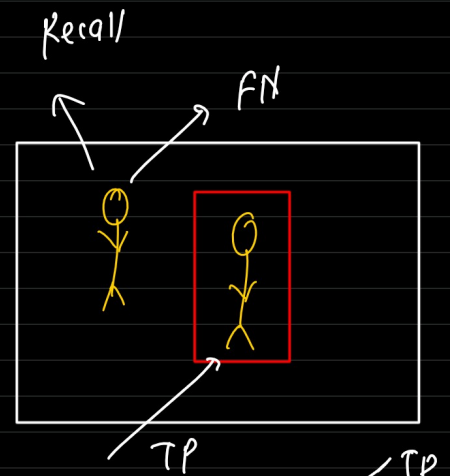
considers true prediction accuracy



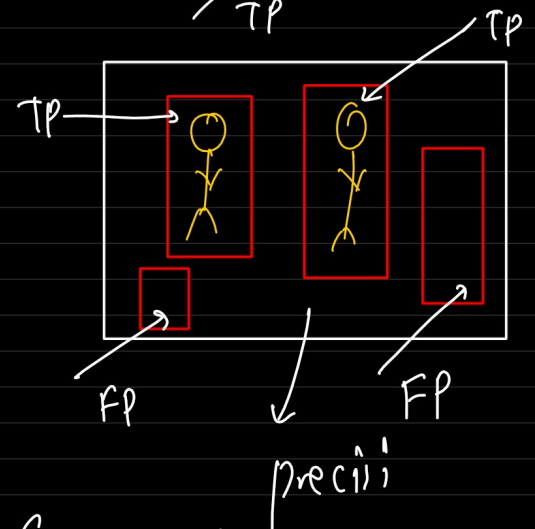
Machine learning algo $\rightarrow$ detect $\rightarrow$ person



$\rightarrow$ ML-1 $\rightarrow$



$\rightarrow$ ML-2 $\rightarrow$



Recall $\rightarrow$ High
precision $\rightarrow$ High

} tradeoff (general)

precision $\rightarrow$ High, Recall $\rightarrow$ low $\rightarrow$ F1 score (P & R)

precision $\rightarrow$ low, Recall $\rightarrow$ High $\rightarrow$ F1 score (P & R)

ML $\rightarrow$ prediction $\rightarrow$ tumor $\rightarrow$

Tradeoff

$> 70\% \rightarrow$ tumor

else $\rightarrow$ non tumor

				<u>70%.</u>	Y
Data $\rightarrow$ 1 $\rightarrow$	ML $\rightarrow$	20%.	F	F	
Data $\rightarrow$ 2 $\rightarrow$	ML $\rightarrow$	75%.	T	T	
$\rightarrow$ 3 $\rightarrow$	ML $\rightarrow$	25%.	F	F	
$\rightarrow$ 4 $\rightarrow$	ML $\rightarrow$	55%.	F $\rightarrow$	T $\rightarrow$ error	
$\rightarrow$ 5 $\rightarrow$	ML $\rightarrow$	80%.	T	T	
$\rightarrow$ 6 $\rightarrow$	ML $\rightarrow$	57%.	F	F	

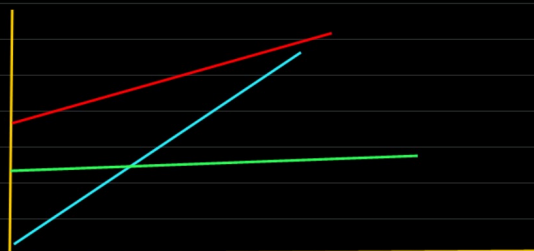
New threshold $\rightarrow > 40\% \rightarrow$ FP $\uparrow$

Equation [Linear, polynomial]

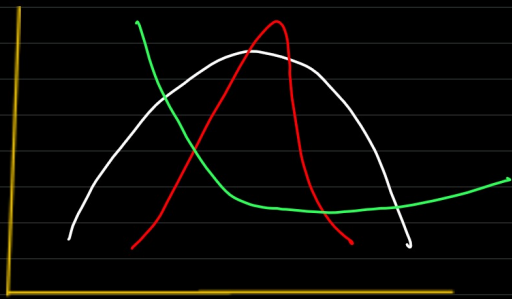
Linear Equation $\rightarrow$ The highest exponent is 1 & their graph are straight lines.

$$\rightarrow y = mx + c \quad \rightarrow 2x - 4y = 8$$

$$\rightarrow y = 3x + 5 \quad \rightarrow f(x) = -x$$



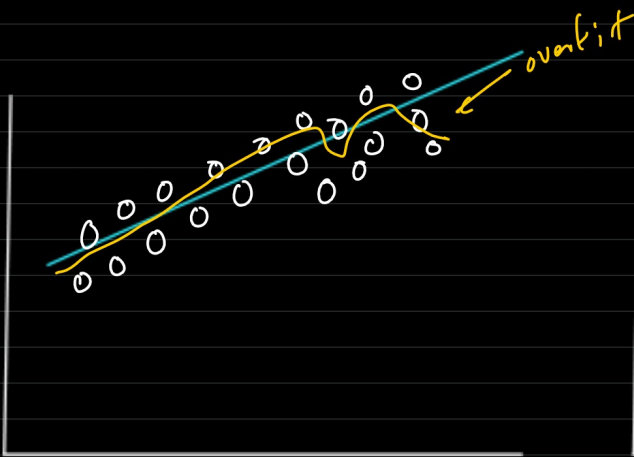
polynomial equation: The highest exponent is 2 or greater (e.g. x^2, x^3, x^n)
 $n \geq 2$ & their graphs are curves with peaks, valleys or U-shapes



→ $y = x^2 + 3x + 2$ (Quadratic)

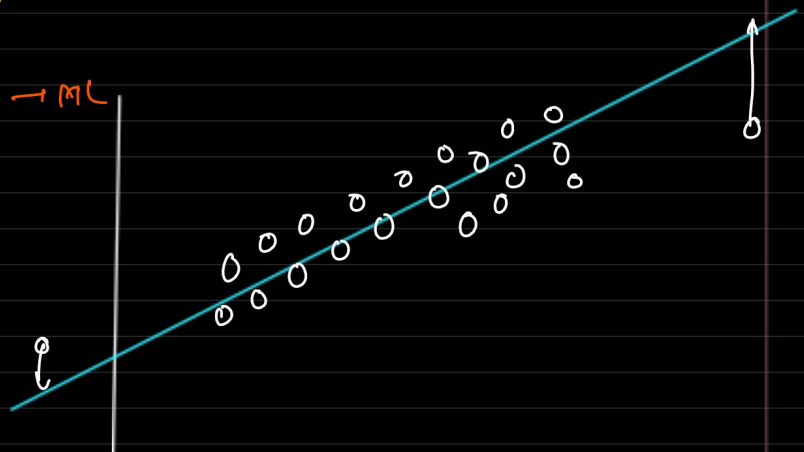
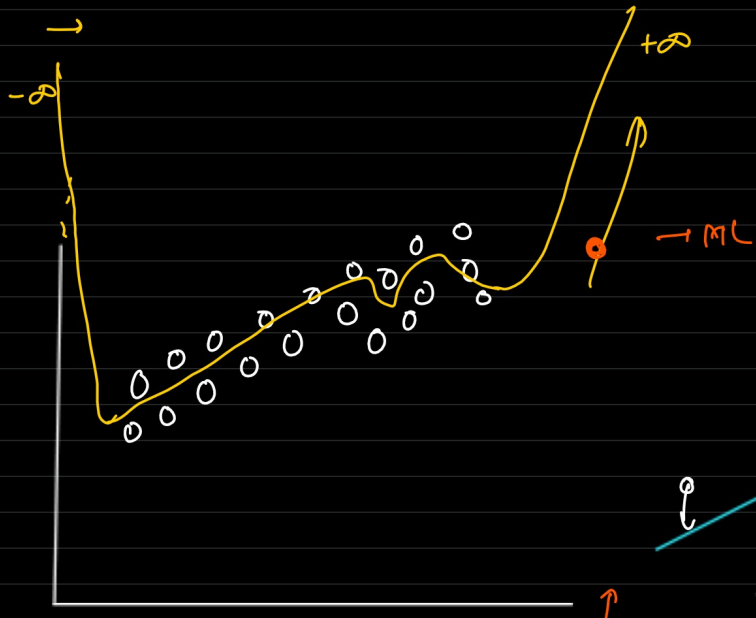
→ $f(x) = x^3 - 4x^2 + x - 5$ (cubic)

→ $y = x^4$ (quartic)

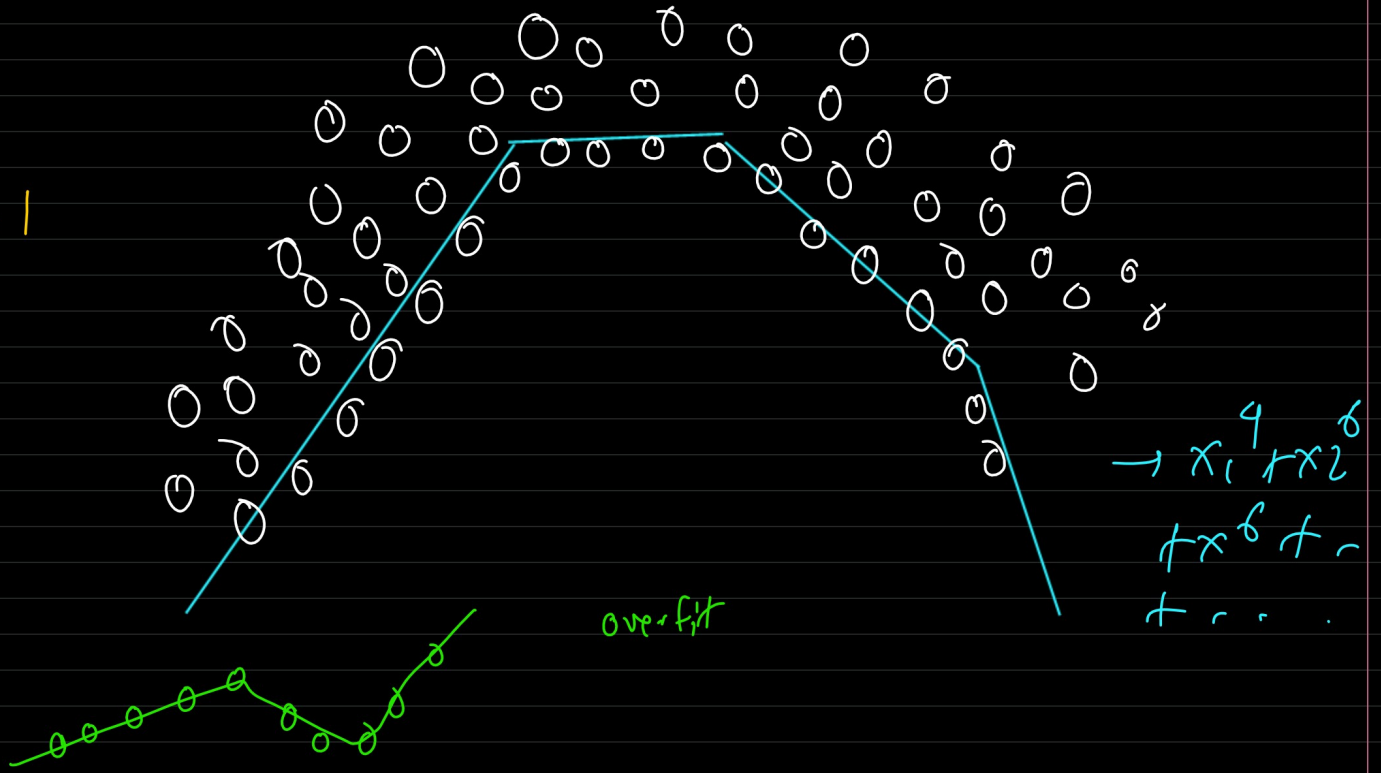


→ linear equation

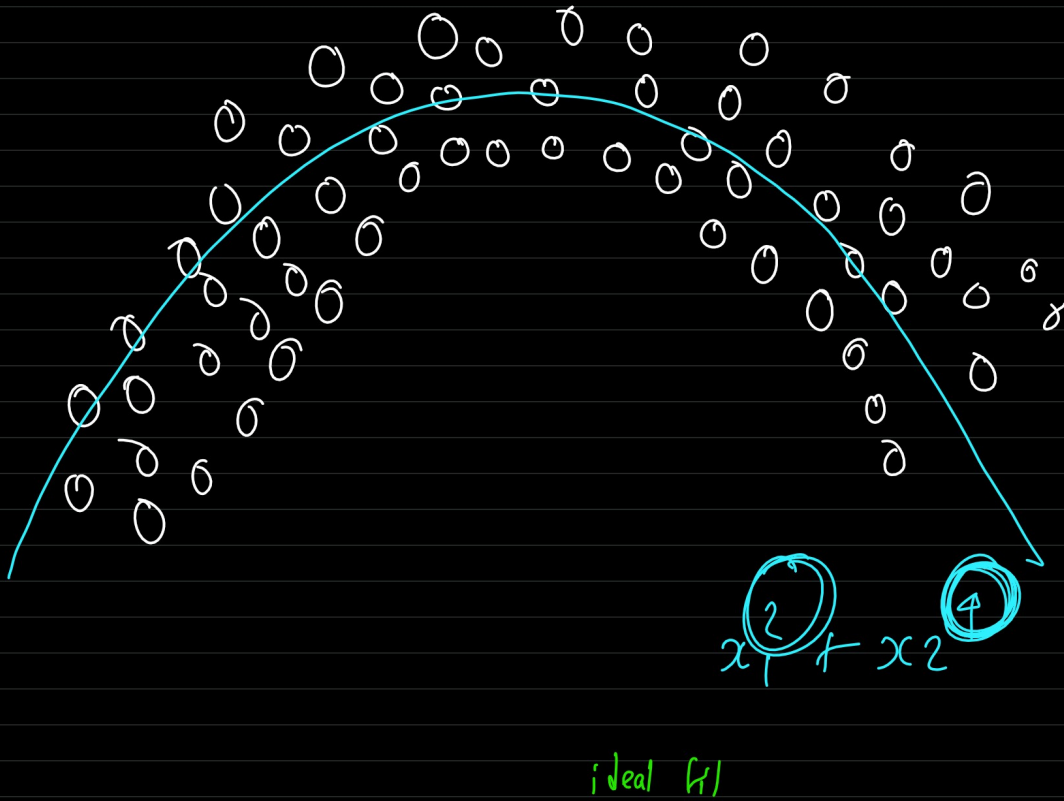
→ polynomial



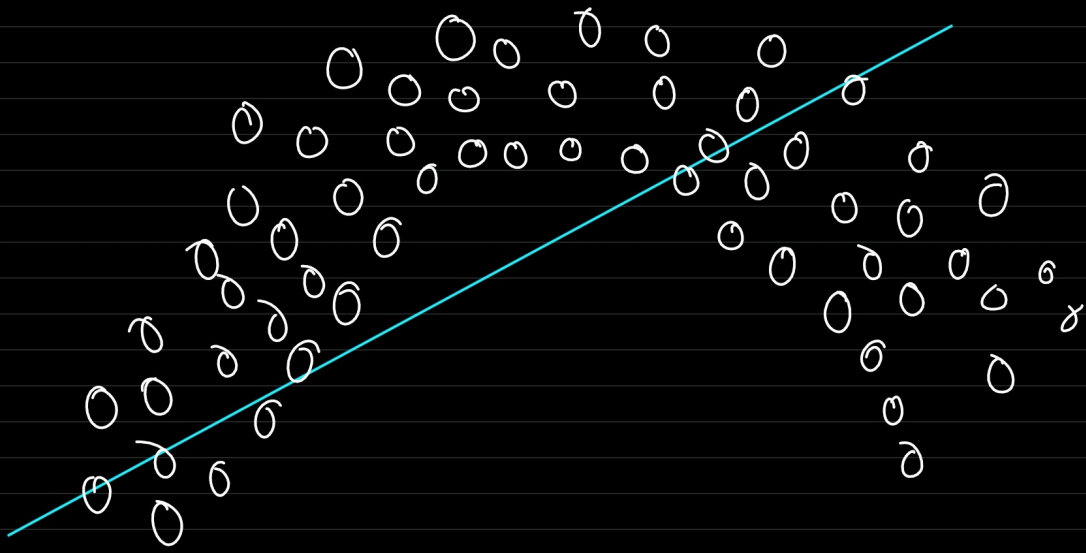
ML-1



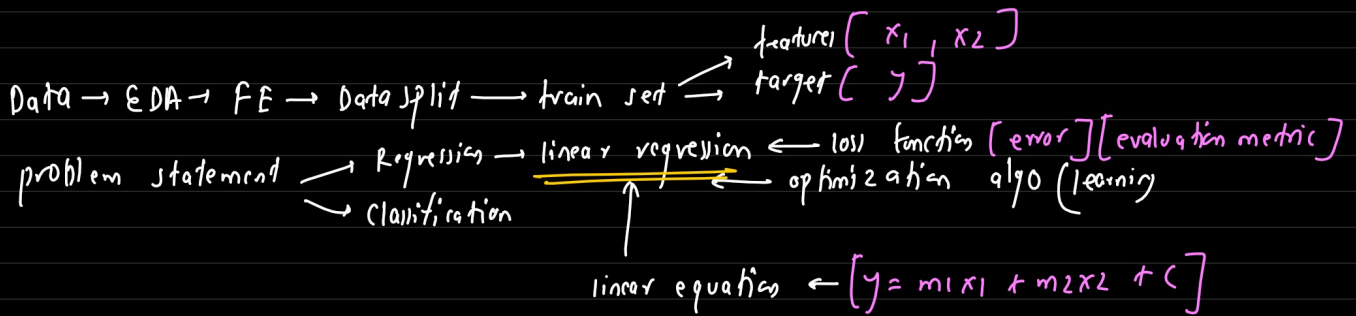
ML-2



ML-3



underfit



x_1	x_2	y
10	20	100
20	30	200
30	40	300

$$\hat{y} \leftarrow m_1 x_1 + m_2 x_2 + c$$

(predicted) $\downarrow$ m_1 m_2 c

$m_1 \rightarrow -10$ $m_2 \rightarrow 10$ $\left| \right.$ $m_1 \rightarrow 10$ $m_2 \rightarrow 0$ $\left| \right.$ ← Trial & error

x_1	x_2	y
0.83	200	149
0.62	192	100
0.72	100	300

→ complex

iteration: 1

data[0] → random (m_1, m_2) → $[1, 1]$ → $x_1 x_1 + x_2 x_1 \rightarrow$

$0.83 \times 1 + 200 \times 1 \rightarrow 200.83$ $\hat{y}$ y error $(\hat{y} - y)$

149 -51.83

data[1] → $\rightarrow$ $\rightarrow$ $\rightarrow$

data[2] → $\rightarrow$ $\rightarrow$ $\rightarrow$

cost function → 1472

iteration 2

$$\text{data}[0] \rightarrow \text{random}(m_1, m_2) \rightarrow [0.8, 0.9] \rightarrow \begin{array}{l} \text{error} \rightarrow \square \\ \text{data}[1] \rightarrow \square \\ \text{data}[2] \rightarrow \square \\ \hline \text{cost function} \rightarrow \boxed{1400} \end{array}$$

$$\begin{array}{l} \text{iteration-1} > \text{iteration-2} \rightarrow m_1, m_2 \downarrow \\ \text{iteration-1} < \text{iteration-2} \rightarrow m_1, m_2 \uparrow \end{array}$$

$$m_1 \rightarrow \uparrow, m_2 \rightarrow \text{const}$$

iteration 3 : $\text{random}(m_1, m_2) \rightarrow \text{cost function} \rightarrow \boxed{1600}$

iteration - 1000 :

$$\begin{array}{cc} m_1 & m_2 \\ \downarrow & \downarrow \end{array}$$

$$\text{cost function} \rightarrow \boxed{85}$$

$$y = x_1 m_1 + x_2 m_2$$

$\downarrow \qquad \downarrow$
optimization

manual optimization $\rightarrow$ minimize the error

$\rightarrow$ we don't know which direction to nudge each parameter (m_1, m_2)

$\rightarrow$ hit & trial

$\rightarrow N$ times

$$x_1 \quad x_2 \quad \dots \quad x_{40}$$

$$m_1 \quad \dots \quad m_{40} \quad + \quad C \quad \rightarrow \text{linear}$$

optimization algo :-

- optimal value of (m_1, m_2) (parameters)
 - ↳ direction to nudge the parameters

$$\begin{array}{ccc} m_1 & m_2 & \\ \downarrow & \downarrow & \\ 1250 & 1220 & + \end{array} \quad \begin{array}{c} \underline{0.0001} \\ \downarrow \end{array} \quad \begin{array}{c} 10,000 \\ \downarrow \\ m_1 \end{array}$$

190 80

→ how much to move / step

$m_1, m_1 \rightarrow \text{error} \rightarrow \approx 0$

$m_1, m_2 \rightarrow \text{error} = \text{large}$

Optimization Algo (Gradient Descent)

Dataset:

Years of exp (x_1)	Completed projects (x_2)	Salary_k (y)
1	0	39.769
2	1	48.900
3	1	56.978
4	2	68.240
5	3	77.867
6	4	85.022

$$\hat{y} = m_1 x_1 + m_2 x_2 + c$$

$$\begin{bmatrix} x_0 & x_1 & x_2 \end{bmatrix}_{n \times 3}$$

1	1	0
1	2	1
1	3	1
1	4	2
1	5	3
1	6	4

$$\begin{bmatrix} m_0 \\ m_1 \\ m_2 \end{bmatrix}_{3 \times 1} \rightarrow \begin{matrix} c \\ m_1 \\ m_2 \end{matrix}$$

$$\begin{bmatrix} x_0 & x_1 & x_2 \end{bmatrix}_{6 \times 3} \begin{bmatrix} m_0 \\ m_1 \\ m_2 \end{bmatrix}_{3 \times 1} \text{ option-1}$$

$$\begin{bmatrix} x_0 & x_1 & x_2 \end{bmatrix}_{6 \times 3} \begin{bmatrix} m_0 & m_1 & m_2 \end{bmatrix}_{1 \times 3} \text{ option-2}$$

$$\begin{bmatrix} x_0 & x_1 & x_2 \end{bmatrix}_{6 \times 3} \begin{bmatrix} m_0 \\ m_1 \\ m_2 \end{bmatrix}_{3 \times 1} \text{ option-1} \rightarrow \text{output shape} \rightarrow 6 \times 1$$

$$\begin{bmatrix} \hat{y} \\ \vdots \\ \hat{y} \end{bmatrix}_{6 \times 1} \begin{bmatrix} y \\ \vdots \\ y \end{bmatrix}_{6 \times 1}$$

$$(\hat{y} - y) = \begin{bmatrix} \vdots \end{bmatrix}_{6 \times 1}$$

$$\text{sum} \rightarrow \begin{bmatrix} \vdots \end{bmatrix}$$

→ back to GD concept

$$\begin{matrix} \circ & \circ & \circ \\ m_1 & m_2 & m_0 \end{matrix} \rightarrow \hat{y} \rightarrow \text{all} \rightarrow 0$$

y	$\hat{y}$	error ($y - \hat{y}$)
39.769	0	39.769
48.400	0	48.400
56.978	0	56.978
68.240	0	68.240
77.867	0	77.867
85.022	0	85.022

cost function (final loss)

$$\sum_{i=1}^n (y - \hat{y}) \rightarrow \sim 376.276$$

$$\rightarrow (m_1 = 0, m_2 = 0, m_0 = 0)$$

$$\rightarrow (m_1 = 0.01, m_2 = 0.1, m_0 = 0.1)$$

→ new final loss →

$$\rightarrow (m_1 = \dots, m_2 = \dots, m_0 = \dots)$$

iteration/set $\rightarrow 1 \rightarrow (m_0 \ m_1 \ m_2) \rightarrow \text{loss} - 1$
 $\rightarrow 2 \rightarrow (m_0 \ m_1 \ m_2) \rightarrow \text{loss} - 2$
 $\rightarrow 3 \rightarrow (m_0 \ m_1 \ m_2) \rightarrow \text{loss} - 3$
 $\rightarrow N \rightarrow (m_0 \ m_1 \ m_2) \rightarrow \text{loss} - N$

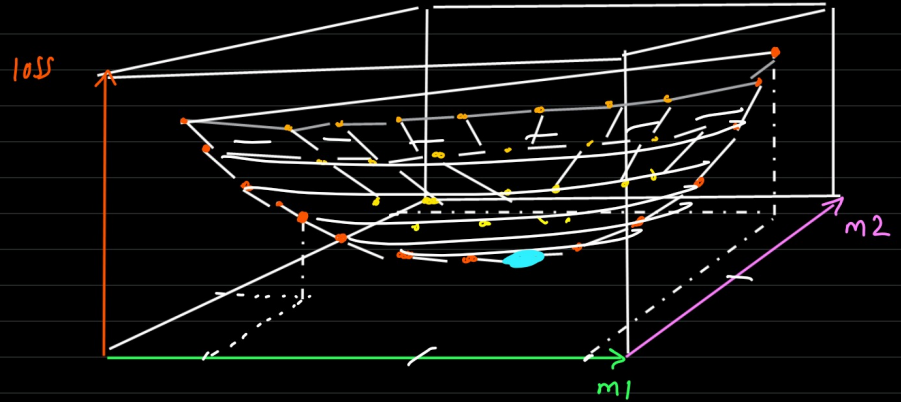
imaginary loss

100

20

400

5



$\rightarrow (1000 \text{ sets}) \rightarrow m_1, m_2 \rightarrow$

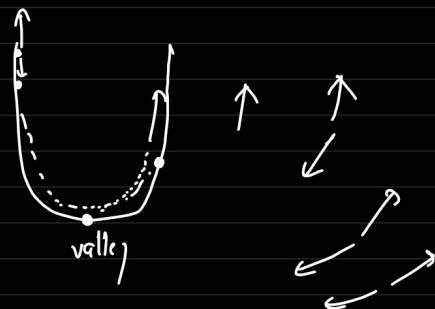
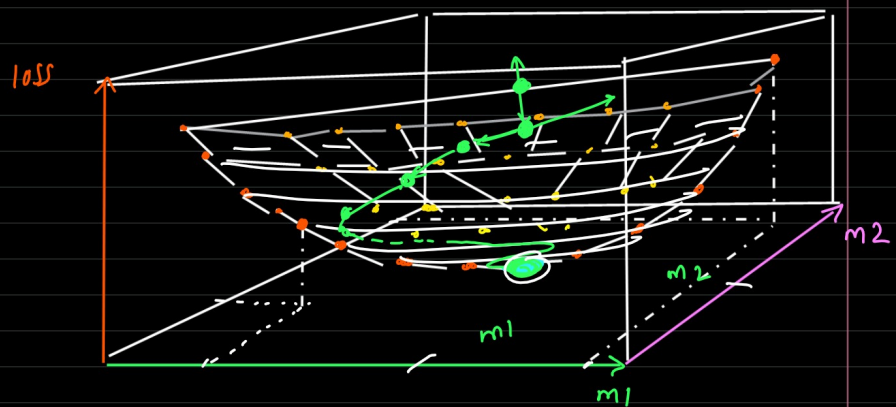
factors for the shape of contour of loss function:

$\rightarrow \lambda$ data

$\rightarrow \text{loss function}$

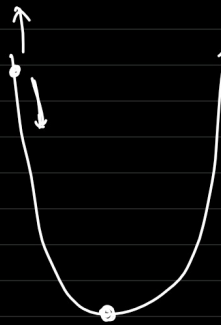
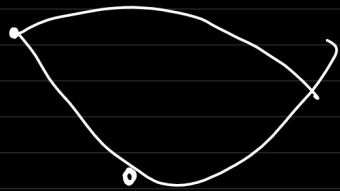
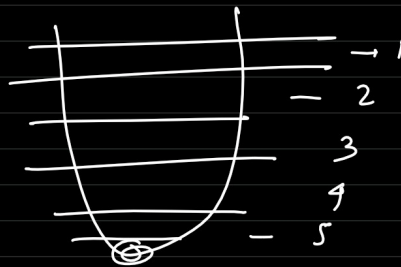
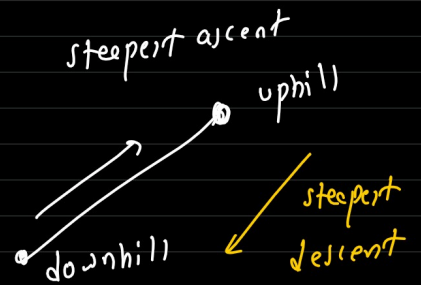
$m_1, m_2 \rightarrow \text{randomly}$

hill $\rightarrow$



Gradient Descent: -

- Goal is to find minimum loss
- use differentiation to calculate slope of the cost function at our current position
- slope/gradient will tell us direction of steepest ascent (uphill)
- Take a step in the opposite direction, steepest descent (downhill)



optimization $\rightarrow$ direction, magnitude $\rightarrow 100$
 $\downarrow$ $\downarrow \times 0.01 \leftarrow$ learning rate

